COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
NAME:	Francis Elijah J. Formalejo	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

```cpp
#include<iostream>
#include <stack>

int main(){
    // create an object:
    std::stack<char> stack1;

    std::cout<<"Is the stack empty? "<<stack1.empty() <<std::endl;
    stack1.push('w');
    std::cout<<"current top of the stack: "<<stack1.top()<<std::endl;
    stack1.push('o');
    std::cout<<"current top of the stack: "<<stack1.top()<<std::endl;
    stack1.push('a');
    std::cout<<"current top of the stack: "<<stack1.top()<<std::endl;
    stack1.push('h');
    std::cout<<"current top of the stack: "<<stack1.top()<<std::endl;
    stack1.push('!');
    std::cout<<"current top of the stack: "<<stack1.top()<<std::endl;
    std::cout<<"Is the stack empty? "<<stack1.empty() <<std::endl;
    std::cout<<"What is the size of the stack? "<<stack1.size() <<std::endl;

    //swap

    //emplace

    return 0;
}
```

```
current top of the stack: w
current top of the stack: o
current top of the stack: a
current top of the stack: h
current top of the stack: !
Is the stack empty? 0
What is the size of the stack? 5
PS C:\Users\TIPQC\Documents\CPE010 -CPE21S4 - FORMLEJO>
```

```cpp
#include<iostream>

//global variables:
#define maxSize 10
int stackArr[maxSize];
int top = -1, newData;

//function prototypes:
bool isEmpty();
bool isFull();
void Top();
void push();
void pop();

int main(){
    int choice;
    while(true){
        std::cout<<"============================\n";
        std::cout<<" Stack operations: " <<std::endl;
        std::cout<<" [1] push [2] pop [3] isEmpty [4] isFull [5] Top"<<std::endl;
        std::cin >> choice;
        std::cout<<"============================\n";
        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: std::cout<<"is the stack empty? "<< isEmpty()<<std::endl;
            break;
            case 4:std::cout<<"is the stack full? "<< isFull()<<std::endl;
            break;
            case 5: Top();
            break;
            default: std::cout<<"Invalid Choice! "<<std::endl;
            break;
        }
    }
    return 0;
}

//function definition
bool isEmpty(){
    return (top == -1);
}

bool isFull(){
    return (top == maxSize-1);
}
```

```
void Top(){
    //error catching:
    if(isEmpty()) return;

    //prints the top element
    std::cout<<"The top element is: "<< stackArr[top]<<std::endl;
}

void push(){
    //error catching:
    if (isFull()) {
        std::cout<<"Stack Overflow!"<<std::endl;
        return;
    }

    //ask the user what they want to insert:
    std::cout<<" What number do you want to push? ";
    std::cin >> newData;

    //storing to the stack:
    stackArr[++top] = newData;

}

void pop(){
    // error catching:
    if(isEmpty()){

        std::cout<<"Stack Underflow!"<<std::endl;

        return;
    }

    // display the item to be popped:
    std::cout << "Popping: " << stackArr[top] << std::endl;

    // we remove the element from the stack
    top--;
}
```

```
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 1
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 1
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 1
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 1
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 1
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Popping: 2
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Stack Underflow!
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
2
============================
Stack Underflow!
============================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top
```

In iloa, I used standard library stack containers because they handle the memories automatically. I avoid some size limits and buffer overflows. In my iloab code, I built a stack from scratch with a ten element array and I set array indicators manually with a top variable, then I wrote code to check for stack overflow and stack underflow.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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